Bodo Handloom Cultural Heritage Platform

Comprehensive System Architecture, Technical Breakdown & Academic Defense Guide

Target: Senior AI Engineer / Research Thesis / CSE Final-Year Review | Date: October 2026

1. Executive Summary & Codebase Reality Check

The Bodo Handloom Cultural Heritage Digital Platform (bodo-handloom-museum) is an interactive virtual museum and cultural preservation system engineered in React 19, TypeScript, Vite, Tailwind CSS, and Three.js. It catalogs sacred Agor motifs, Ahimsa Eri silk heritage, botanical dyeing workflows, and 3D loom implements of the Bodo indigenous community of Northeast India.

Critical Engineering Discovery: The repository is currently a Production-Grade Frontend Application with in-browser visual CMS and WebGL 3D virtual village. The backend REST APIs, LangGraph multi-agent retrieval orchestrator, and Vector/Graph databases are currently defined as Architectural Contracts in frontend/src/services/api.ts and frontend/src/components/ai/ChatPanel.tsx. There are no Python files or running vector databases yet in the repository; the RAG pipeline is stubbed and awaiting backend implementation.

2. Complete Repository File & Directory Breakdown

Core Configuration and Server Entrypoints

- package.json & server.ts: Root package configuration and Node.js production HTTP server listening on 0.0.0.0 with Cloud Run health checks (/api/health) and SPA fallback.
- frontend/package.json & vite.config.ts: Configures Vite 8 with React 19 and Tailwind CSS v4 compiler plugin, serving port 3000.
- metadata.json: Application metadata defining identity, capabilities, and cloud run deployment flags.

Application Core & Component Hierarchy

- frontend/src/main.tsx & App.tsx: React application entrypoint. Initializes global navigation, renders cultural sections, handles modals, and listens to view routing.
- frontend/src/services/api.ts: The critical boundary file. Defines AIQueryRequest and AIQueryResponse interfaces. Throws a deliberate pending-integration error to prevent ungrounded mock answers.
- frontend/src/components/ai/: Contains ChatPanel, ChatHeader, ChatInput, ChatMessage, ExpertiseSelector, and SuggestedQuestions. Implements educational UX with 4 expertise levels (Basic, Intermediate, Advanced, Expert).

- frontend/src/components/3d/village/: Complete Three.js village environment including VillageBuilder.ts (6 cultural zones), VillageAssetSystem.ts, Insert3DObjectModal.tsx, and ObjectPropertiesPanel.tsx.
- frontend/src/store/: Zustand stores managing state: museumStore.ts (modal/filter state), contentStore.ts (in-browser CMS), villageAssetStore.ts, and villageObjectStore.ts (transform persistence).
- frontend/src/data/: Domain registries: motifs.ts (sacred Agor patterns), dyeing.ts (botanical dye recipes), weavers.ts (master artisans), artifacts.ts, and processes.ts.

3. The RAG & AI Pipeline: Reality vs. Implementation Plan

For a final-year thesis or research paper, the distinction between what is currently written in code and what is conceptually specified must be presented with absolute honesty:

- Current Status: Frontend interface is 100% built. It captures user query, selected expertise level, and timestamps in React state and localStorage. It logs payload to console and renders an educational pending-connection card.
- Next Phase (Your Responsibility): Build the FastAPI / Python backend with LangGraph. Integrate Bodo heritage document ingestion (PDFs of ethnographic surveys, textile manuals, botanical records), chunking, and hybrid retrieval.
- Hybrid Retrieval Architecture: BM25 (sparse keyword match for indigenous terms like 'Dokhona', 'Aronai', 'Siju', 'Gourd') combined with Dense Vector Retrieval (Sentence-Transformers / BGE embeddings) via Reciprocal Rank Fusion (RRF).
- GraphRAG Integration: Connecting entities (Weaver -> Village -> Motif -> Dye Plant -> Cultural Ceremony) in Neo4j to resolve multi-hop queries that pure vector search fails on.

4. Research Methodology & Genuine Academic Research Gaps

As your Research Professor, here are the three high-impact, publishable research gaps suitable for a final-year BTech paper or IEEE/ACM workshop submission:

- Research Gap 1 - Low-Resource Indigenous Lexicon Degradation in Dense Embeddings: Standard multilingual models (e.g. text-embedding-3-small, mBERT) exhibit catastrophic semantic collapse on Sino-Tibetan and Bodo terminology. A hybrid sparse-dense retriever with domain-specific BM25 boosting provides measurable recall gains.
- Research Gap 2 - Expertise-Conditioned RAG Generation: Generating answers tailored to visitor persona (child/enthusiast vs academic textile researcher) while preventing factual hallucination of sacred motifs.
- Research Gap 3 - Multi-Hop Cultural Graph Traversal: Answering complex queries such as 'Which botanical dyes used in Dokhona Thaosi are harvested during Baisagu festivals?' requires traversing knowledge graph edges rather than single document chunks.

5. Evaluation Metrics & Experimental Framework

To defend this project scientifically in front of evaluators, use the RAG Triad framework:

- Context Relevance: Precision@K and MRR (Mean Reciprocal Rank) measuring whether retrieved PDF chunks actually contain the true Bodo historical facts.
- Groundedness / Faithfulness: Measuring the fraction of generated statements directly supported by retrieved evidence (detecting hallucinations).
- Answer Relevance: Semantic similarity between generated output and verified master weaver oral interview transcripts.

6. Folder Organization for PDFs & Research Materials

A dedicated directory /pdf has been created at the root of the project, with a mirror in frontend/public/pdf for live browser access. Store your source documents here using the following structure:

- pdf/research/: Academic papers, field ethnographic surveys, and botanical chemistry publications.
- pdf/dyeing/: Botanical dye recipes, extraction manuals, mordant dosage charts, and plant field notes.
- pdf/motifs/: Agor geometric diagrams, mathematical grid patterns, and symbolic meaning documentation.
- pdf/weavers/: Transcripts of recorded weaver oral histories, guild records, and award citations.

7. Final-Year Viva Voce Defense Questions & Answers

- Q: Why use Hybrid Search instead of pure Vector Search? A: Indigenous terms (like 'Dokhona Thaosi', 'Zenther', 'Maku') are out-of-vocabulary for pre-trained vector models. BM25 guarantees exact keyword hits while vector embeddings capture conceptual meaning.
- Q: Why is Three.js used client-side instead of server rendering? A: Three.js WebGL runs with zero server latency directly in the client GPU, supporting 60 FPS walking navigation, model insertion, and transform editing.
- Q: How is data persisted currently? A: Browser LocalStorage (contentStore.ts and villageObjectStore.ts) and IndexedDB (modelStorage.ts for 3D GLB binaries), providing instant persistence without cloud dependencies.